

An injectable self-healing hydrogel-cellulose nanocrystals conjugate with excellent mechanical strength and good biocompatibility

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ABSTRACT

In this work, a novel strategy for the construction of injectable self-healing nanocomposite (NC) hydrogels dominated by reversible boronic ester bonds was demonstrated. Specifically, NC hydrogels were constructed by the solution-mixing of *N,N*-dimethylacrylamide-*stat*-3-acrylamidophenylboronic acid statistical copolymers (PDMA-*stat*-PAPBA) and poly(glycerolmonomethacrylate) (PGMA) chains grafted cellulose nanocrystals (CNC-g-PGMA). Rheology analysis indicated the as-constructed NC hydrogel displayed about 7-fold increase in the storage modulus with a low CNCs loading level of 1.43 wt% in comparison with PGMA/PDMA-*stat*-PAPBA hydrogel without CNCs. Furthermore, the mechanical strength of the CNC-g-PGMA/PDMA-*stat*-PAPBA hydrogel was far superior to that of its PGMA/PDMA-*stat*-PAPBA/CNCs hydrogel counterpart, in which PGMA chains were not covalently grafted on the surfaces of CNCs. Due to reversible boronic ester bonds cross-linking networks, CNC-g-PGMA/PDMA-*stat*-PAPBA NC hydrogel exhibited excellent self-healing and injectable properties as well as pH/glucose responsive sol-gel transitions. Good biocompatibility was also demonstrated through *in vitro* cytotoxicity tests.

1. Introduction

Hydrogels are three-dimensional cross-linked polymeric networks containing a great deal of water (Fan, Wang, & Jin, 2018; Shen, Shamshina, Berton, Gurau, & Rogers, 2016). Because of their high water content and solid-like properties, hydrogels have gained wide applications ranging from tissue engineering, wound repair to drug delivery, etc (Culver, Clegg, & Peppas, 2017; Hamed, Moradi, Hudson, & Tonelli, 2018; Li, Song et al., 2018; Reakasame & Boccaccini, 2017). However, the conventional hydrogels cross-linked by covalent chemical bonds are much brittle due to the lack of effective energy dissipation mechanism, (Cong, Wang, & Yu, 2014; Xu, Li, Wang, Zhang, & Zhang, 2013). This inherent deficiency of chemical gels largely limits their application scopes. As a result, the construction of tough hydrogels with good mechanical properties, such as double-network hydrogels, nanocomposite (NC) hydrogels, supermolecular hydrogels, etc, have received considerable interest in the past decades (Cheng et al., 2018; Li, Jiang, & Haraguchi, 2018; Li, Yang, Zhao, Long, & Zheng, 2017;

Nascimento et al., 2018; Sun, Wang, & Yan, 2017; Wang et al., 2018). Among which, NC hydrogels by the noncovalent or covalent immobilization of nanoparticles in hydrogel matrix have received special attention on account of many available functional nanoparticles. Diverse kinds of nanoparticles, such as metal, metal oxide, and inorganic as well as polymeric nanoparticles, have been utilized to prepare hydrogel-nanoparticle conjugates (Thoniyot, Tan, Karim, Young, & Loh, 2015). The combination of nanoparticles with hydrogels have created many synergistic and unique properties that are unattainable for the individual components (Thoniyot et al., 2015). As one of the most promising classes of the bio-materials, hydrogels are usually implanted into the body to act as drug carriers or physical supports for cell culture. In this regard, those hydrogels simultaneously exhibiting self-healing, injectable and stimuli-responsive properties demonstrate many merits compared with chemical hydrogels (Ding et al., 2010). Firstly, hydrogels implanted in the body may undergo deformation or damage by the external mechanical force, which will increase the risk of infection and thus shorten their life spans. The autonomous healing themselves after

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damage can enhance the safety in the body and reduce the replacement cost (Liu et al., 2015). Secondly, injectable hydrogels can improve the adaptation to the target defect site during surgical implantation (Yu & Ding, 2008). Lastly, stimuli-responsive sol-gel transition of hydrogels allows bioactive molecules such as drugs and proteins to be easily loaded and released on demand at target site *via* external stimuli (Tomatsu, Peng, & Kros, 2011). Hydrogels holding simultaneously self-healing, injectable, and stimuli-responsive properties are commonly constructed either by supramolecular interactions or reversible covalent bonds (Wei et al., 2014). Regrettably, till today, practical applications of self-healing hydrogels are rarely found mainly due to poor mechanical properties.

Cellulose nanocrystals (CNCs) are highly crystalline nanorods with a width varying from 5 to 20 nm and lengths from 100 nm to 2000 nm (Habibi, Lucia, & Rojas, 2010). Due to high axial elastic modulus and tensile strength, rod-like CNCs are excellent candidates for fillers to reinforce the strength and stiffness of composite materials (Moon, Martini, Nairn, Simonsen, & Youngblood, 2011). Recently, a series of NC hydrogels exhibiting excellent toughness and flexibility have been constructed by the incorporation of CNCs into hydrogel matrix *via in-situ* radical polymerization method (Yang, Han et al., 2013; Yang et al., 2012; Zhou, Wu, Yue, & Zhang, 2011). Numerous reactive hydroxyl groups on the surfaces of CNCs provide versatile strategies for the post-modification of CNCs. Benefiting from this, functional groups or polymeric chains could be easily grafted onto the surfaces of CNCs through different chemical processes (Alizadehgiashi et al., 2018; Bai, Huang, Xie, & Xiong, 2018; De France, Chan, Cranston, & Hoare, 2016; Li, Jiang et al., 2018; Liu et al., 2018; Shao et al., 2019; Shao, Wang, Chang, Xu, & Yang, 2017; Shao et al., 2018; Yang, Bakaic, Hoare, & Cranston, 2013). By the employment of functionalized CNCs as both reinforcing fillers and cross-linkers, a series of self-healable NC hydrogels with good mechanical properties have been constructed (Alizadehgiashi et al., 2018; Bai et al., 2018; De France et al., 2016; Li, Zhang, Wu, Guo, & Luo, 2018; Liu et al., 2018; Shao et al., 2019, 2017; Shao et al., 2018; Yang, Bakaic et al., 2013). The homogeneous dispersion of reinforcing fillers in gel matrix can create good surface adhesion and large interfacial area, which is certainly crucial to their reinforcement performance and the improvement of universal properties of the resulting NC hydrogel (Alam, Zhang, Kuan, Lee, & Ma, 2018; Yang, Han, Zhang, Xu, & Sun, 2014). In the case of CNCs, however, a part of sulfate groups are unavoidably removed from the surfaces of CNCs during post-modification (Jiang, Esker, & Roman, 2010; Roman & Winter, 2004), which deteriorates the dispersion of functionalized CNCs in water.

In this study, we demonstrated a novel strategy for the construction of self-healable hydrogel-CNCs conjugate. Different from the previous strategies for the construction of hydrogels containing CNCs or functionalized CNCs (Alizadehgiashi et al., 2018; Bai et al., 2018; De France et al., 2016; De France, Hoare, & Cranston, 2017; Li, Zhang et al., 2018; Liu et al., 2018; Shao et al., 2019, 2017; Shao et al., 2018; Yang, Han et al., 2013; Yang et al., 2012; Yang, Bakaic et al., 2013; Zhou et al., 2011), NC hydrogels were achieved by the solution-mixing of the hydrophilic diol-containing polymers grafted CNCs and phenylboronic acid-containing binary copolymers. The resultant hydrogel networks were dominated by reversible boronic ester bonds. Due to the integration of the stimuli-responsiveness of physical gels, such as pH/sugar responsiveness and the stability of chemical gels, reversible boronic ester bonds cross-linking hydrogels have been intensively studied in recent years (Chen, Diaz-Dussan et al., 2018; Chen, Tan, Wang, Peng, & Narain, 2018; Chen, Wang et al., 2018; Deng, Brooks, Abboud, & Sumerlin, 2015; Guan & Zhang, 2013; Guo et al., 2017; Lu et al., 2016; Piest, Zhang, Trinidad, & Engbersen, 2011; Smithmyer et al., 2018). However, to the best of our knowledge, no studies like the current method have been revealed. The specific objective was to demonstrate the significant reinforcement performance of CNCs resulting from the covalent graft of a polymeric component.

2. Experimental

2.1. Materials

CNCs and 3-acrylamidophenylboronic acid (APBA) were prepared according to our early reports (Du, Guo, Li, & Gao, 2017), respectively. *N,N*-Dimethylacrylamide (DMA) was purchased from Aladdin Industrial Corporation (China), which were firstly passed through a basic Al_2O_3 column and then distilled under the reduced pressure prior to polymerization. Glycerol monomethacrylate (GMA) was purchased from Shanghai G&K Biomedical Scientific Inc (China), which was firstly passed through a basic Al_2O_3 column prior to polymerization. Ammonium persulfate (APS) was purchased from Aladdin Industrial Corporation (China), which was purified by recrystallization method. All other reagents were obtained from commercial suppliers and used without further purification.

2.2. Synthesis of CNC-g-PGMA graft copolymers

Ammonium persulfate (320.00 mg, 1.40 mmol) was added to the round-bottom flask containing 132 mL of CNCs aqueous dispersion (20 mg/mL). The flask was degassed with nitrogen for 1 h. The flask was sealed and then transferred to an oil bath at 60 °C. After 15 min, the temperature was decreased to 50 °C, and 1 mL of degassed GMA aqueous solution (11.80 mmol/mL) was injected to the flask. The temperature was raised rapidly to 60 °C. After 10 h of polymerization, the reaction was stopped by exposure to air, followed by cooling the reaction mixture to room temperature. The solution was concentrated, and solid samples were separated by centrifuge. The solid samples were washed with deionized water. The dispersion, separation and washing procedures were conducted repeatedly to remove the adsorbed PGMA homopolymers as possibly. The resulting solid samples were re-dispersed in deionized water. After lyophilization, 3.81 g of fluffy white solid was afforded. The relative content of PGMA (RC_{PGMA}) grafted on the surface of CNCs was roughly calculated to be ~30 wt% according to gravimetric method using the following equation:

$$\text{RC}_{\text{PGMA}} = (W_1 - W_0)/W_1$$

Where RC_{PGMA} , W_1 , and W_0 were the relative content of the PGMA chains grafted on the surfaces of CNCs, and the mass of CNC-g-PGMA copolymers and the mass of CNCs, respectively. The mass loss of the samples during the transfer was neglected.

2.3. Synthesis of PDMA-stat-PAPBA copolymers

DMA (3.30 g, 33.20 mmol), APBA (3.17 g, 16.60 mmol), AIBN (17.80 mg, 0.10 mmol), and 10 mL of fresh dimethylacetamide (DMAc) were charged in a round-bottom flask equipped with a magnetic stirrer. After being degassed with nitrogen for 1 h, the flask was sealed and then transferred to an oil bath at 70 °C. After 8 h of polymerization, the reaction was terminated by the addition of methanol. The mixture was precipitated into a large amount of cold diethyl ether. The precipitation was collected and washed with cold diethyl ether for three times. And the purified product was dried under vacuum at room temperature. The mole fractions of DMA and APBA units in the copolymers were analyzed by ^1H NMR spectroscopy. ^1H NMR (400 MHz, CD_3OD , δ , ppm): 8–7 (benzene ring protons), 3.2–2.8 ($-\text{N}(\text{CH}_3)_2$) (Fig. S1); ^{13}C NMR (100 MHz, CD_3OD , δ , ppm): 33–47 ppm (the backbone of polymer carbons and methyl carbons); 123–139 ppm (phenyl carbons); 176 ppm ($\text{C}=\text{O}$ carbons) (Fig. S2). GPC analysis (DMF eluent, refractive index detector): $M_n = 23,800$ g/mol and $\text{PDI} = 1.77$ (Fig. S1), and a series of near-monodisperse of polystyrene samples were used as standards.

2.4. Preparation of CNC-g-PGMA/PDMA-stat-PAPBA NC hydrogels

A typical procedure for the preparation of CNC-g-PGMA/PDMA-stat-PAPBA NC hydrogel as follows: 30.00 mg of CNC-g-PGMA copolymers was dispersed in 1.00 mL of alkaline water at pH 10. And then, 0.30 mL of alkaline aqueous solution (pH 10) containing 70 mg of PDMA-stat-PAPBA was added. The gelation occurred rapidly under the stirring. Other hydrogels with different compositions were prepared with the same procedures described above. The content of water was fixed to be ~93 wt% for all hydrogels.

2.5. Self-healing of CNC-g-PGMA/PDMA-stat-PAPBA NC hydrogels

Two hydrogel disks with the same compositions were prepared by the above-mentioned method, and one of disk was stained by Rhodamine B (RhB) dye. The hydrogels were cut into two pieces, respectively. And two different colored semicircles were put together in the original round bottle without any external intervention at room temperature.

2.6. pH- and glucose-triggered gel-sol transitions

In order to assure the molecular dissolution of PDMA-stat-PAPBA copolymers during sol-gel transition, DMF/water binary solvents were used to prepare NC hydrogel. The volume ratio of DMF/H₂O was 3:7. For a performed nanocomposite hydrogel, 15 μ L of HCl solution (1 mol/L) was added to adjust pH of the original NC hydrogel from 10 to 5. The NC hydrogel was dissociated into sol under the vigorous shaking. For regelation, 20 μ L of NaOH aqueous solution (1 mol/L) was added to the above sol accompanying vigorous shaking. The cyclic gel-sol transitions were performed by the alternate addition of HCl and NaOH.

For the glucose triggered gel-sol transition, nanocomposite hydrogels were immersed in PBS (pH 7.4) buffer with or without glucose. The concentration of glucose in PBS buffer was 6 mg/mL. Hydrogels were dyed with RhB for a better observation.

2.7. In vitro cytotoxicity tests

The human umbilical vein endothelial cells (HUVECs; purchased from ScienCell Research Laboratories, Carlsbad, CA) were cultured in endothelial cell medium as described elsewhere (Ji et al., 2017). HUVECs were seeded at a density of 4×10^4 /well on 24-well plates and grown for 2 days prior to exposure. THP-1 cells (ATCC, Manassas, VA, USA) were cultured in RPMI 1640 medium (Thermo-Fisher, NY, USA) supplemented with fetal bovine serum and penicillin-streptomycin solution and differentiated into macrophages by the treatment of phorbol 12-myristate 13-acetate (PMA; Sigma, St. Louis, MO, USA) as previously described (Jiang et al., 2016). The cells were seeded at a density of 2.4×10^5 /well on 24-well plates prior to exposure. NC hydrogel was treated for 8 min with continuous cooling on ice by an ultrasonic processor FS-250N (20% amplitude; Shanghai Sheng xi, China) and then diluted in cell culture medium as 1.00%, 0.50%, 0.25%, 0.13% and 0.07%, in which the concentration was the mass fraction of polymer components including CNC-g-PGMA and PDMA-stat-PAPBA in the mixture. The cells were incubated with various concentrations of NC hydrogels or 1% Triton-X (as positive control) for 24 h, and then neutral red uptake assay was done to indicate cytotoxicity by using commercial kits according to manufacturer's instruction (Beyotime, Nantong, China). The products were read by an ELISA reader (Synergy HT, BioTek, Woburn, MA, USA). In vitro cytotoxicity tests for S2 and S3 were conducted according to the similar procedures as described above, and the concentration of hydrogel was 1.00%.

2.8. Characterization and test

¹H NMR spectra of polymers and ¹³C NMR spectrum of PDMA-stat-

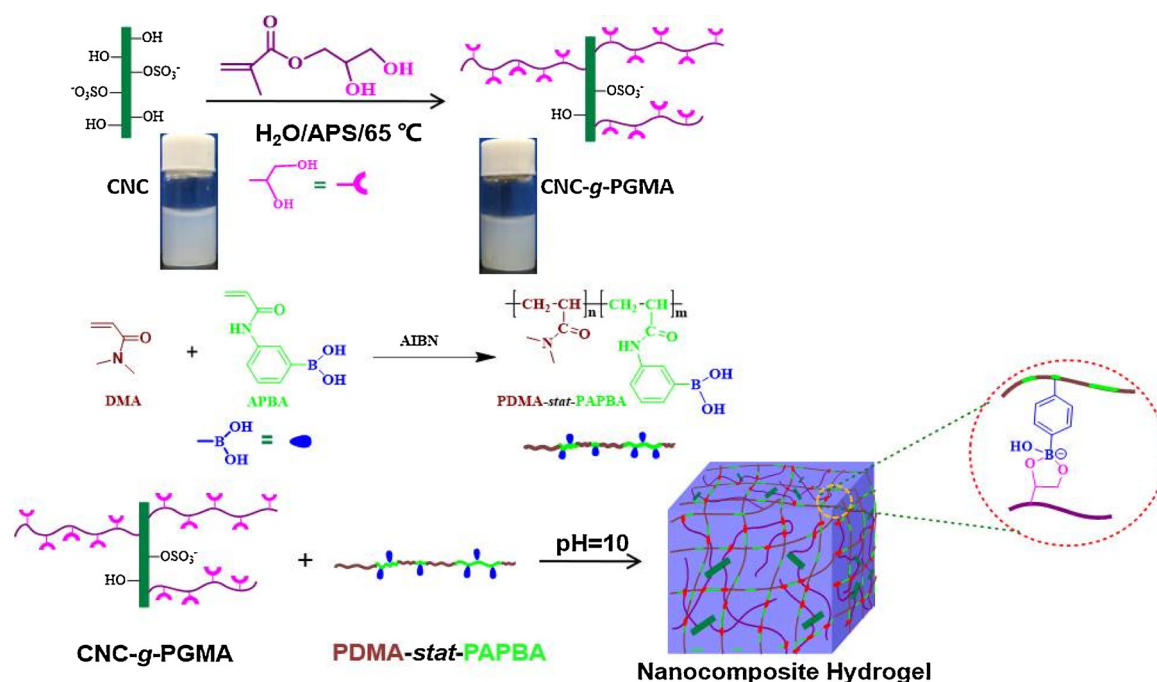
PAPBA were recorded on a Bruker AV-400 NMR spectrometer. The solid-state CP/MAS (cross-polarization-magic angle spinning) ¹³C NMR spectra were recorded at room temperature with a Bruker Avance DSX400 MHz spectrometer at a magic angle spinning (MAS) rate of 6 kHz, and at a frequency of 75.5 MHz. The sample was packed in MAS4-mm-diameter zirconia rotors. All spectra were run for 7000 scans. Number-average molecular weights (M_n) and the molecular weight distribution of copolymers were determined by GPC measurements, which were performed on a Waters 1515 GPC equipped with a Waters 2414 differential refractive index detector in DMF at 80 °C with a flow rate of 1.0 mL/min. FT-IR spectra were recorded on a Perkin-Elmer Spectrum One FT-IR spectrometer by the KBr pellet method at a resolution of 2 cm⁻¹ in the range of 4000 cm⁻¹–400 cm⁻¹. Thermo gravimetric analysis (TGA) was performed on a TA SDT Q600 instrument under a nitrogen atmosphere at a heating rate of 10 °C/min in the range from 30 to 550 °C. Elemental analysis (EA) was conducted with a Perkin-Elmer CHN 2400 analyzer. Zeta potential measurement was performed on a MALVERN Zetasizer Nano ZS instrument (Malvern Instruments Ltd, Malvern, UK). Each sample was measured for three times and the average value was calculated. Crystalline structures of CNCs and CNC-g-PGMA were studied on a diffractometer (D8 advance XRD, Bruker Company, Germany) using Cu K α radiation (λ = 0.154 nm, 40 kV and 40 mV) from 2 θ = 10–40° at a scan rate of 4°/min. X-ray photoelectron spectroscopy (XPS) spectra were obtained on VG Escalab MKII instrument with an Mg K α X-ray source (15 keV, filament current of 20 mA). All binding energies were referenced to C 1s hydrocarbon peak at 284.6 eV. A Shirley function was used to correct the background of all spectra. Rheological measurements were performed by using an AR2000Ex (TA Instruments) rheometer with a parallel plate geometry (20 mm diameter acrylic plate). All rheological measurements were conducted at room temperature.

3. Results and discussion

3.1. Synthesis of CNC-g-PGMA graft copolymers

CNCs were obtained from sulfuric acid hydrolysis, which have been characterized carefully in our earlier study (Du et al., 2017). In this study, the same batch of CNCs materials was used. Therefore, morphological characterizations, including TEM and AFM were not re-conducted. The average length, width and the thickness of CNCs were 84 ± 23 nm, 4.9 ± 0.6 nm, 5.4 ± 0.9 nm, respectively (Fig. S3). CNC-g-PGMA copolymers were synthesized by “graft from” strategy and APS was used as initiators. The specific route was illustrated in Scheme 1, and the same procedure was reported in early study (Tang et al., 2014). During graft copolymerization, a small amount of PGMA originated from the solution phase polymerization was inevitable, which was removed as possibly through post-treating operations including separation and washing. The resulting CNC-g-PGMA could be well dispersed in water at a concentration of 30 mg/mL, as shown in Scheme 1.

CNC-g-PGMA was carefully characterized by FT-IR, EA, XPS, TGA and ¹³C NMR as well as XRD. Typical FT-IR spectrum of CNC-g-PGMA was displayed in Fig. S4. Compared with FT-IR spectrum of CNCs, an obvious absorption peak at ~1730 cm⁻¹, corresponding to the characteristic stretching vibration of the ester groups of PGMA chains, was observed, implying the successful graft of PGMA chains onto the surface of CNCs. EA results of the pristine CNCs and CNC-g-PGMA samples were demonstrated in Table S1. After the graft of PGMA, the contents of C and H elements of CNCs increased as expected, whereas S element content decreased from 0.6% to 0.4%. The reduced sulfur content was ascribed to the partial removal of the sulfate groups from the surfaces of CNCs during the chemical reaction (Jiang et al., 2010; Roman & Winter, 2004). The partial removal of the sulfate groups was also supported by the zeta potential measurement (Table S1). In spite of the partial removal of sulfate groups from the surfaces of CNCs, CNC-g-PGMA could



Scheme 1. The schematic illustration of synthetic routes for CNC-g-PGMA, PDMA-*stat*-PAPAB and CNC-g-PGMA/PDMA-*stat*-PAPAB NC hydrogel; The concentrations of the aqueous dispersions were 20 mg/mL for CNCs and 30 mg/mL for CNC-g-PGMA, respectively.

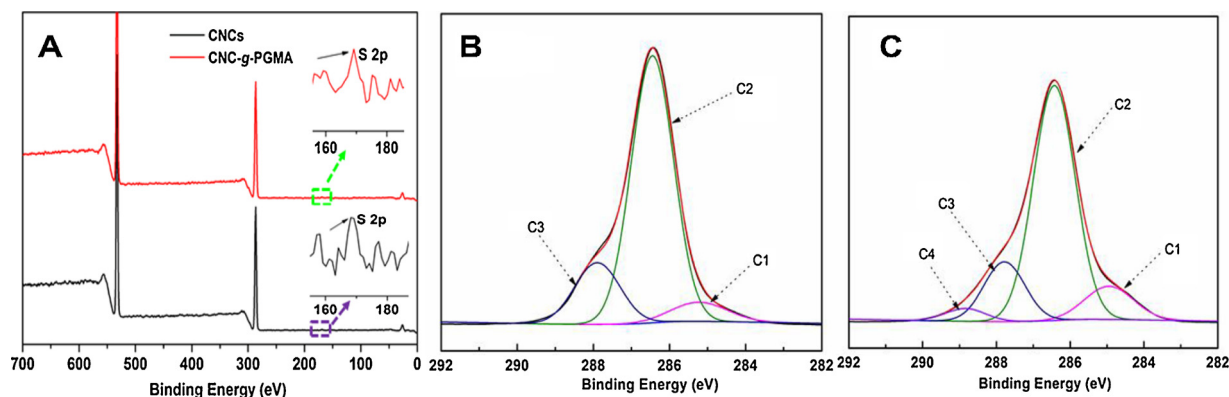


Fig. 1. XPS survey spectra of CNCs and CNC-g-PGMA (A); High-resolution C 1s scans for CNCs (B) and CNC-g-PGMA (C).

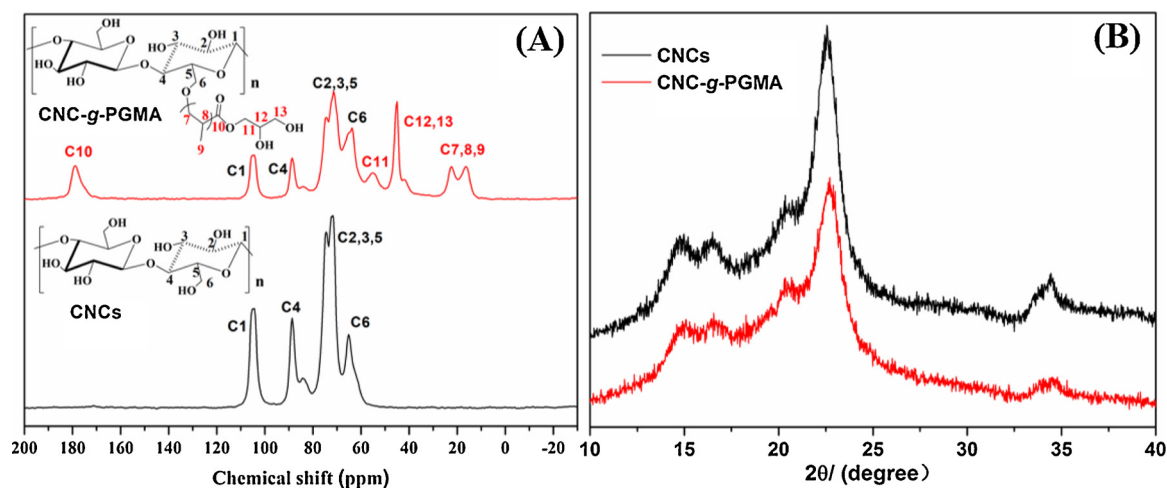


Fig. 2. (A) ¹³C NMR spectra of CNCs and CNC-g-PGMA; (B) XRD patterns of CNCs and CNC-g-PGMA.

Table 1
Shear rheology results of hydrogels with various compositions.

Sample ^a	CNCs (wt%)	PGMA (wt%)	PDMA- <i>stat</i> -PAPBA (wt%)	PBA/Diol ^c (molar ratio)	G' (Pa)	ω_c (rad/s)
S1 ^d	1.43 ^b	0.71 ^b	5.00	2.50	~62,000	–
S2	0.00	0.87	6.12	2.50	~93,000	0.1
S3 ^e	1.43	0.71	5.00	2.50	~24,000	0.31

^a The water content in all samples was fixed to be ~93 wt%.

^b the relative mass fractions of CNC and PGMA in S1 were roughly calculated based on the RC_{PGMA}.

^c The total mole number of PBA groups was roughly calculated based on the mass of PDMA-*stat*-PAPBA and the molecular weight as well as the relative mole fraction of APBA in copolymer; the total mole number of diol groups were roughly calculated based on the mass of PGMA.

^d PGMA chains were grafted on the surfaces of CNCs in S1.

^e PGMA chains were not grafted on the surfaces of CNCs in S3.

still be well dispersed in water owing to the graft of the hydrophilic PGMA chains. The graft of PGMA on the surfaces of CNCs was evidenced by XPS spectroscopy. XPS survey spectra of CNCs and CNC-g-PGMA were indicated in Fig. 1A. Two obvious peaks at 284.6 eV and 532.4 eV, corresponding to C 1s and O 1s, respectively, were observed in the respective survey spectrum. A discernable weak peak at 169.8 eV was attributed to S 2p, revealing the presence of the sulfate groups on the surfaces of CNCs and CNC-g-PGMA. O/C ratio decreased from ~0.78 for CNCs to ~0.73 for CNC-g-PGMA as the result of the grafting of carbon-rich PGMA chains. C 1s peak of CNCs was curve-fitted to three different carbon environments at 284.8 eV, 286.5 eV and 288.1 eV, corresponding to C–C/C–H, C–O and C–O–C species, respectively, as shown in Fig. 1B. C 1s peak of CNC-g-PGMA could be decomposed four different types of carbon bonds at 284.8 eV, 286.5 eV and 288.1 eV as well as 289.1 eV (Fig. 1C), which were assigned to C–C/C–H, C–O, C–O–C and O–C=O species, respectively (de Paula et al., 2016; Zhang et al., 2017). O–C=O specie was attributed to the ester groups of the grafted PGMA chains. The relative content of C–C/C–H component increased from 16.2% for CNCs to 17.1% for CNC-g-PGMA, whereas the relative content of C–O component decreased from 60.1% to 49.7%.

The solid-state ¹³C NMR spectra of the original CNCs and CNC-g-PGMA were shown in Fig. 2A. In the ¹³C NMR spectrum of the original CNCs, the resonance at 65 ppm was attributed to C6, the primary alcohol group of the anhydroglucose units. The resonances appearing at 68–80 ppm corresponded to C2, C3, and C5 of anhydroglucose units. The resonances at around 105 ppm and 89 ppm were assigned to C1 and C4 of anhydroglucose units, respectively (Fig. 2A) (Spinella et al., 2016; Wu et al., 2015). Compared with that of the original CNCs, the solid-state ¹³C NMR spectrum of CNC-g-PGMA exhibited several groups of new resonances at 178 ppm, 55 ppm, 45 ppm and 11–26 ppm, which was attributed to the corresponding carbons of PGMA chains, as indicated in Fig. 2A. The solid-state ¹³C NMR data was in favor of the covalent linkage of PGMA chains with CNCs.

X-ray diffraction was utilized to investigate the crystalline structures of the pristine CNCs and CNC-g-PGMA samples. The XRD pattern of the pristine CNCs was illustrated in Fig. 2B. Five groups of diffraction peaks appearing at 2 θ around 15.1°, 16.6°, 20.8°, 23.0° and 34.8°, were assigned to (1–10), (110), (102), (200) and (004), respectively (Duchemin, Le Corre, Leray, Dufresne, & Staiger, 2016; Novo, Bras, García, Belgacem, & Curvelo, 2015). These peaks did not display any shift after the grafting of PGMA chains, indicating the intact lattice constants of CNCs during the graft polymerization. This also suggested that all PGMA chains were grafted on the surface of CNCs, and the internal hydroxyl groups did not take part in the graft polymerization reaction. TGA/DTG traces of CNCs, PGMA and CNC-g-PGMA samples were displayed in Fig. S5. PGMA was completely decomposed at the temperature ranging from ~200 °C to ~500 °C, as illustrated in Fig. S5 (trace a). CNCs showed ~38 wt% mass loss at the temperature ranging from ~180 °C to ~325 °C, and the weight loss was attributed to the dehydration of some glucose units on the main chain of CNCs, and the breakage of the molecular backbone as well as the breakage of other C–O and C–C bonds (Bano & Negi, 2017). ~22 wt% of mass loss at the

temperature ranging from ~325 °C to ~420 °C was the result of the formation of carbon. The final residue was ~31 wt% (Fig. S5, trace b). CNC-g-PGMA exhibited a two-stage thermal degradation process. The first thermal degradation was found at temperature ranging from ~240 °C to ~340 °C, corresponding to ~50 wt% of weight loss. The second thermal degradation was found at the temperature ranging from ~340 °C to ~450 °C, corresponding to ~30 wt% of weight loss (Fig. S5, trace c), and the final residue was ~20 wt%. Compared with CNCs, CNC-g-PGMA showed an increased initial decomposition temperature, revealing the improved thermal stability. This might be attributed to the partial removal of sulfate groups from the surface of CNCs (Roman & Winter, 2004).

3.2. Preparation and mechanical properties of NC hydrogels

Boronic ester cross-linking NC hydrogel networks were constructed by the mixing of CNC-g-PGMA and PDMA-*stat*-PAPAB copolymers in solution at pH10 (Scheme 1). PDMA-*stat*-PAPAB were synthesized by traditional free radical copolymerization, whose structure, molecular weight and the distribution of molecular weight were characterized by ¹H NMR and GPC, and the results were given in Fig. S1. According to ¹H NMR analysis, the average molar fraction of APBA unit in PDMA-*stat*-PAPAB copolymers was ~33%. The compositions of various hydrogels were summarized in Table 1. The molar ratio of PBA/diol in all hydrogels was fixed to be 2.5.

Commonly, nanoparticles would associate/interact with each other by virtue of the strong interactions (such as hydrogen bonding) in NC hydrogels if the loading level of the nanoparticles reaches the critical concentration, such as percolation threshold (ν_{RC}). In such case, tridimensional percolating network would be formed (Dufresne, 2008; Lin & Dufresne, 2013), which is helpful for the strengthening and improvement of the mechanical properties of the resulting NC hydrogels. For CNCs nanoparticles, the value of ν_{RC} can be evaluated according to the following equation (Lin & Dufresne, 2013):

$$\nu_{RC} = \frac{0.7}{L/d}$$

L and d were the length and the diameter of CNCs, respectively. L/d represented the aspect ratio. On the basis of TEM observation (Fig. S2), ν_{RC} for the current system was calculated to be ~4.16 vol%, corresponding to ~6.66 wt% referring 1.6 g/cm³ of the density of CNCs. However, the content of CNCs in S1 (Table 1) was far lower than 6.66 wt%, and the association/interaction between CNCs nanoparticles was excluded. Gel networks of S1 were mainly dominated by the reversible boronic ester bonds formed by PAB/diol complexation interaction in alkali surroundings. The neighboring CNCs were bridged by PDMA-*stat*-PAPAB/PGMA linkages, leading to a homogeneous dispersion of CNCs in the gel matrix. CNCs acted as both the cross-linkers and the reinforcing fillers in gel matrix. Thus, there existed dual cross-linking effects in the NC hydrogel networks, as illustrated in Scheme 1. The dependences of the storage modulus (G') and the loss modulus (G'') of the constructed NC hydrogels on the frequency at a constant strain of

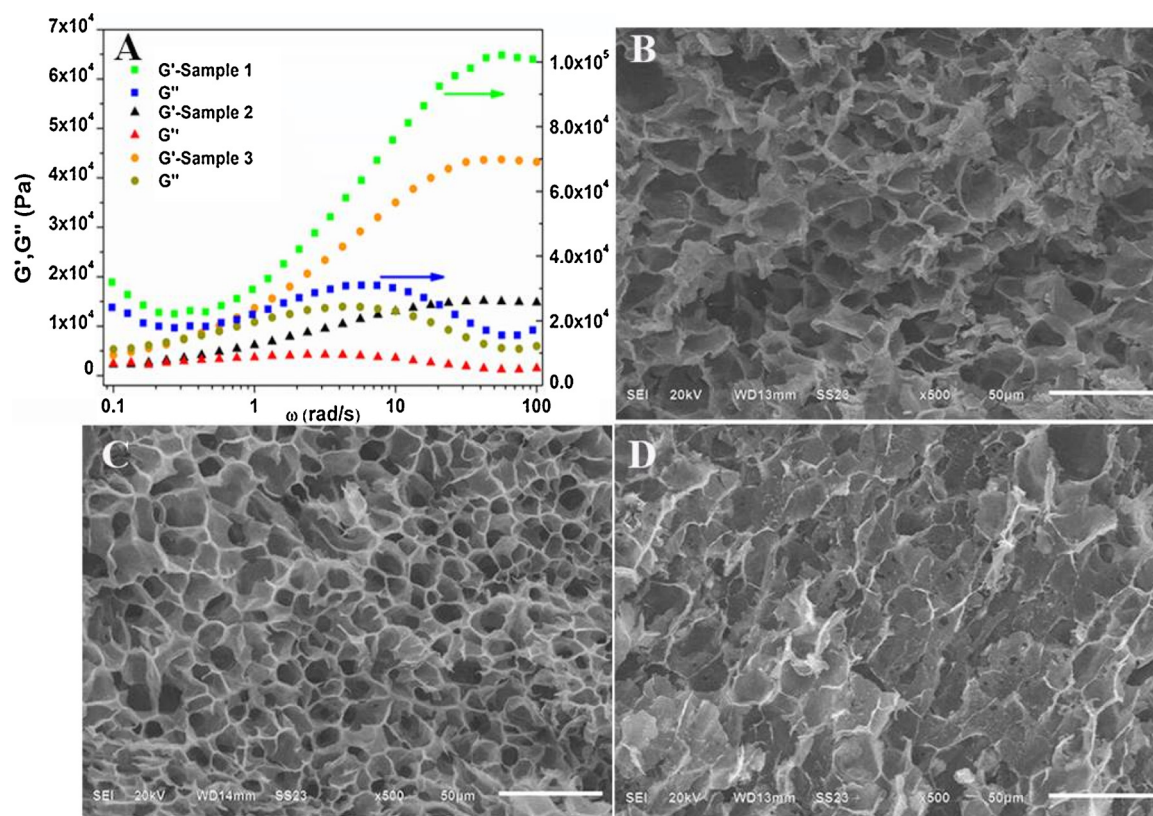


Fig. 3. (A) G' and G'' of S1, S2 and S3 as functions of the frequency at a constant strain of 1%; (B–D) Typical SEM images of the lyophilized S1, S2 and S3 samples.

1% were shown in Fig. 3. As indicated in Fig. 3A, NC hydrogel of S1 exhibited frequency-dependent viscoelastic behavior, revealing the typical dynamic hydrogels networks (Chen, Wang et al., 2018; Lin & Dufresne, 2013). At low frequencies, hydrogels behaved like liquid with $G'' > G'$, whereas G' exceeded G'' at high frequencies, indicating a solid-like gel state. The crossover frequency (ω_c), at which G' is equal to G'' , is usually applied to evaluate the gel strength. Hydrogels with lower ω_c are considered to be more elastic because the mechanical properties are dominated by G' over a wider range of angular frequency (Piest et al., 2011). G' of S1 was $\sim 61,700$ Pa, and no ω_c was detected within the angular frequency ranging from 0.1 to 100 rad/s, whereas G' and ω_c of S2 without CNCs were $\sim 9,300$ Pa and ~ 0.11 rad/s, respectively. G' of S1 was ~ 6 times as much as that of S2, indicating the significant reinforcement effect of CNCs. NC hydrogel of S3 was constructed by the direct mixing of CNCs, PDMA-*stat*-PAPBA, and PGMA in solution at pH10, which was described as PDMA-*stat*-PAPAB/PGMA/CNCs (Table 1). The content of each component of S3 was the almost same as that of S1. G' and ω_c were $\sim 24,300$ Pa and 0.32 rad/s on the basis of rheology measurements, respectively. Evidently, the mechanical strength of S1 was far superior to that of S3. This difference should be the result of the different interfacial adhesion between the reinforcing fillers of CNCs and the gel matrix. The interaction between CNCs and gel matrix in S3 was weak hydrogen bond, whereas it was covalent linkage in S1. The covalent bond energy is commonly much higher than that of hydrogen bond. A strong interfacial adhesion rendered an additional increase in the number of elastically active chains per unit of volume (Yang et al., 2014). The microstructures of the three hydrogels were observed with SEM. Fig. 3B–D showed the typical SEM images of the lyophilized S1, S2 and S3 samples. All the hydrogels exhibited heterogeneous porous microstructures, indicating the typical polymer-based hydrogel structures. Among the three hydrogels, S2 exhibited the smallest pore size, and NC hydrogels of S1 and S3 formed relative large pores based on SEM observation. For the same molar ratio of diol/PBA, the increased pore size suggested the reduced cross-linking density in

gel networks, which could be inferred that the complexation interaction between PBA groups and diol groups was interfered to some extent by CNCs due to the hindrance effect.

3.3. Self-healing and injectable properties

The self-healing behavior of S1 was investigated. Two pieces of semicircular gel plates with red color (dyed with RB) and light-yellow color (original gel) were intimately contacted (Fig. 4A–C). It was found two semicircles with different colors were completely merged into an integral hydrogel disk within ~ 20 min (Fig. 4D). The fractured interface became blurred. At the side of light-yellow half, a light red color was observed next to the boundary (Fig. 4D), which suggesting that the formation of boronic ester across the damaged interface allowed the gel to be healed. In other words, the self-healing of the NC hydrogel was the result of both the mobility of polymer chains and the reversibility of boronic ester bonds (Amaral, Emamzadeh, & Pasparakis, 2018; Cash, Kubo, Bapat, & Sumerlin, 2015). The healed hydrogel was strong enough to be stretched by tweezers without any rupture at the boundary (Fig. 4E, F).

The self-healing behavior of NC hydrogel of S1 was further investigated with continuous step strain sweeps. Prior to this measurement, the dependence of moduli on the strain at a constant frequency of 1 rad/s was studied and the results were indicated in Fig. 5A. As indicated, the result showed that both G' and G'' remained constant within $\sim 10\%$ of the strain. With the increase of the applied strain, both G' and G'' showed a significant drop, indicating a quasi-liquid state. The crossover was observed at $\sim 320\%$ of the strain. Beyond this critical strain point, G'' was greater than G' , implying a sol state. The continuous step strain sweeps were shown in Fig. 5B. G' was largely dropped from $\sim 61,000$ K Pa to ~ 1300 Pa upon the application of 400% of strain. When the strain was returned to 1.0% , G' and G'' of S1 were almost recovered to the initial levels rapidly. For example, G' was $\sim 59,000$ Pa at 1% of the strain, $\sim 97\%$ of the original level. This recovery

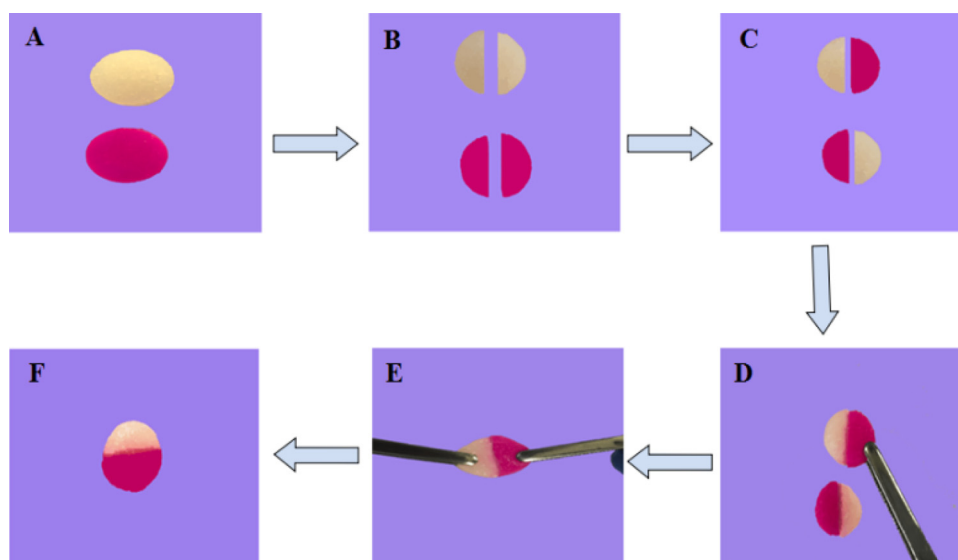


Fig. 4. Demonstration of the self-healing of NC hydrogel of S1 at room temperature.

behavior was repeatable for at least three cycles. NC hydrogels of S1 exhibited shear-thinning behavior (Fig. 5C), revealing the tendency of rod-like particles to align with the flow field (Wu, Jiang, Zan, Lin, & Wang, 2017; Zhang et al., 2010). Fig. 5C (inset) demonstrated the typical injectability of S1 through a 22-gauge needle. This injectability was resulted from the synergetic effects between the alignment tendency of CNCs with the flow field and the fast dissociation/reformation of the reversible boronic ester bonds (Tang et al., 2018; Zhi et al., 2018).

3.4. pH- and glucose-responsive gel-sol transition

The current NC gel networks were mainly dominated by the reversible boronic ester bonds. Because of pH- and glucose-responsiveness of the boronic ester bond (Guo et al., 2017), the responsive gel-sol transition could be achieved not only by the change of pH, but also by

the addition of glucose. Fig. 6 (Panel A) indicated pH-triggered reversible gel-sol transition. Upon the addition of 15 μ L of HCl (1 mol/L) into the system, gel was rapidly turned to sol because of the dissociation of PBA/diol complexes. This could be interpreted that most of PBA would exist in the form of the uncharged trigonal states and lost the capability of complexation with *cis*-1,2-diol when pH was lower than pKa of PBA. After pH was adjusted back to the initial state by the addition of NaOH aqueous solution (1 mol/L), gel was regenerated. This reversible gel-sol phase transition could be carried out for several cycles.

Sugar-triggered disassociation was conducted by the immersing of S1 into PBS buffer (pH 7.4) containing glucose. PBS buffer without glucose was used as a control. The concentration of glucose was 6 mg/mL. NC hydrogel of S1 was dyed with Rh B for a better observation. The changes were recorded at specific time intervals. It was observed that a small proportion of hydrogel was disassociated after 1 h of standing,

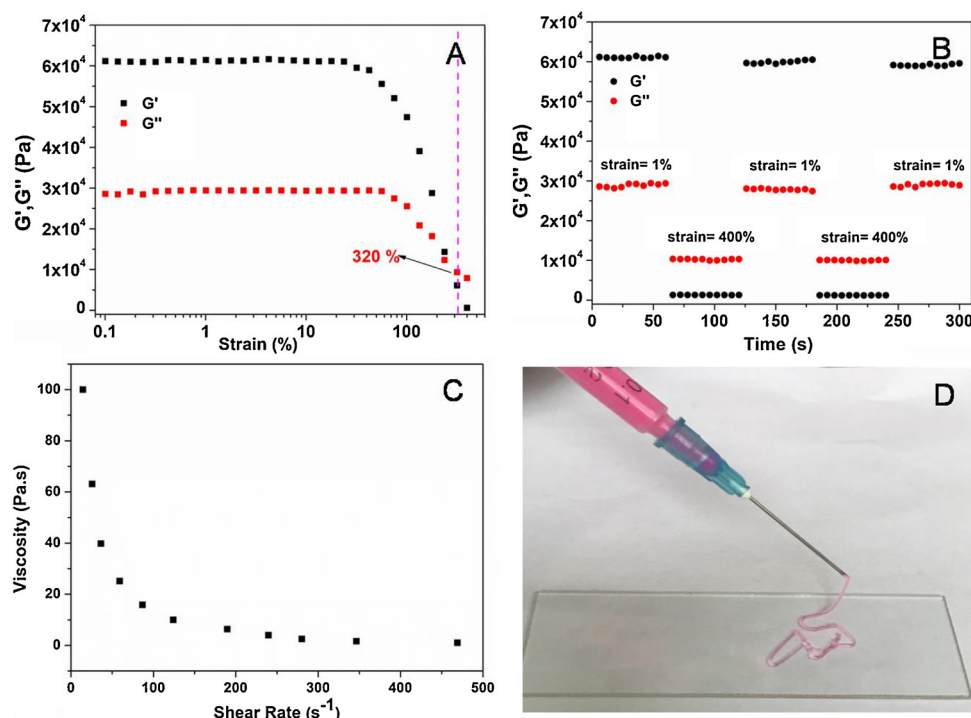


Fig. 5. (A) Dependence of moduli of NC hydrogel of S1 on strain amplitude sweep ($\gamma = 0.1\% - 400\%$) at a fixed frequency of 1 rad/s; (B) Step-strain test of NC hydrogel of S1 at a fixed frequency of 1 rad/s (1% or 400% of strain was applied); (C) Viscosity measurement of NC hydrogel of S1 at 1% of strain; (D) Demonstration of injectability of S1 dyed with RB through a 22-gauge needle.

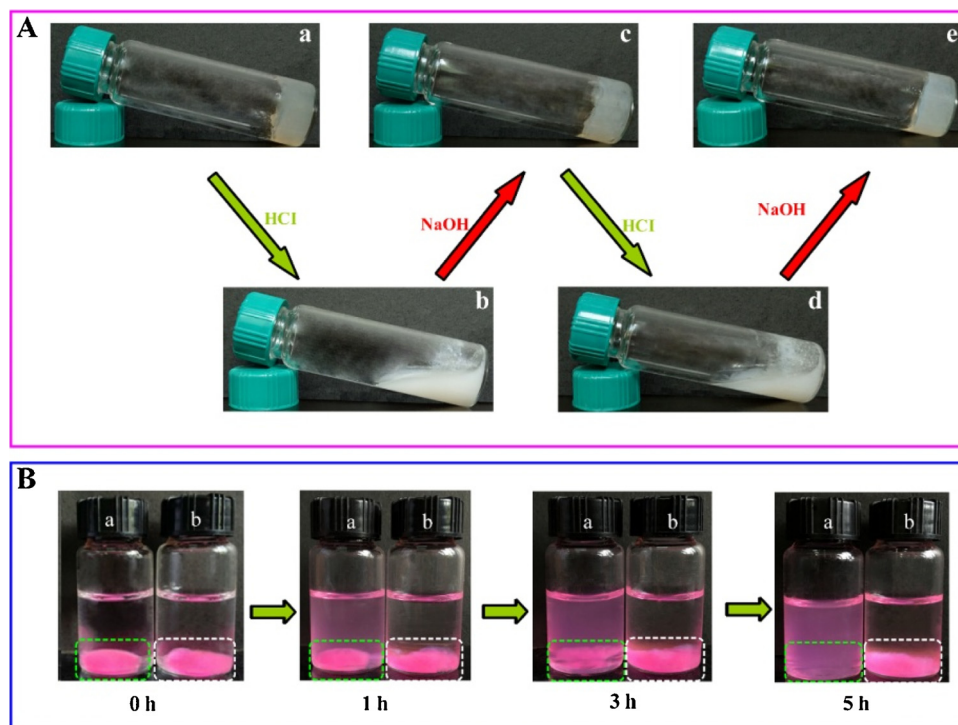


Fig. 6. (A) pH-triggered reversible gel-sol transition of NC hydrogel of S1; (B) Glucose-triggered disassociation of NC hydrogel of S1.

and complete disappearance was observed within 5 h. However, there was no obvious change for S1 immersed in PBS solution without glucose over the time scale of observation. All these were shown in Fig. 6 (Panel B). The disassociation of S1 was derived from the competitive complexation with PBA between GMA units and glucoses. The complexation capability of glucose with *cis*-1,2-diol with PBA is stronger than that of GMA, leading to a gradual replacement of PDMA-*stat*-PAPAB/PGMA complexes by PBA/glucose complexes. As a result, the gel networks were disrupted.

3.5. In vitro cytotoxicity tests

Cell cytotoxicity tests were performed by incubating THP-1 macrophages or HUVECs with culture medium containing NC hydrogel for 24 h. Exposure to various concentrations of S1 did not significantly decrease the viability of THP-1 macrophages (Fig. 7A) or HUVECs (Fig. 7B), which indicated NC hydrogel of S1 has good biocompatibility, with a relative cell viability of more than 90%. Similar experimental

results were found for both S2 and S3 with 1% of hydrogel concentration (Fig. S6). Such excellent cytocompatibility endowed the constructed NC hydrogels with promising application as a biomedical material.

4. Conclusion

Hydrophilic PGMA chains were successfully grafted on the surfaces of CNCs by “graft from” strategy. By the mixing of CNC-g-PGMA and PDMA-*stat*-PAPBA in solution, an injectable self-healable NC hydrogel with good mechanical strength was constructed. CNCs were homogeneously dispersed in the hydrogel matrix and acted as both the reinforcing fillers and the cross-linkers. Compared with PGMA/PDMA-*stat*-PAPBA hydrogel, the mechanical strength of the constructed NC hydrogels of CNC-g-PGMA/PDMA-*stat*-PAPBA gained significant improvement at a low content of CNCs. Additionally, the mechanical strength of as-constructed CNC-g-PGMA/PDMA-*stat*-PAPBA NC hydrogels was far superior to that of its counterpart of PDMA-*stat*-PAPBA/

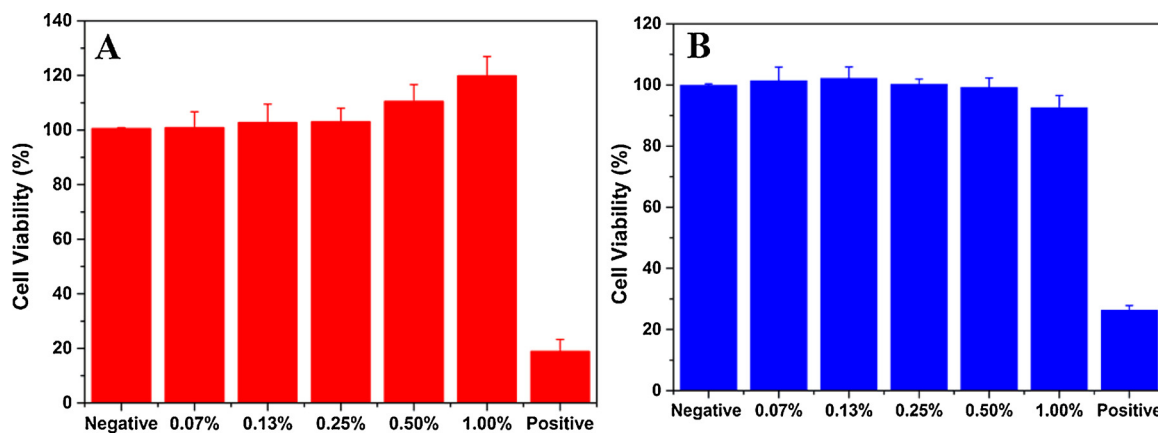


Fig. 7. The cytotoxicity tests of NC hydrogel of S1 to THP-1 macrophages (A) or HUVECs (B); The cells were exposed to various concentrations of S1 or 1% Triton X (positive control) for 24 h. Neutral red uptake assays were done to indicate cytotoxicity.

PGMA/CNCs because of covalent linkage between the reinforcing fillers of CNCs and hydrogel matrix. The CNC-g-PGMA/DMA-*stat*-PAPBA NC hydrogel exhibited perfect self-healing behavior, good injectability, and pH/sugar dual-responsive sol-gel transitions owing to reversible boronic ester bonds dominated gel networks. Besides, CNC-g-PGMA/DMA-*stat*-PAPBA NC hydrogel also exhibited good biocompatibility. All these rendered CNC-g-PGMA/PDMA-*stat*-PAPBA NC hydrogel with promising potentials in drug delivery and tissue engineering, etc.

Declaration of Competing Interest

The authors disclose no potential conflicts of interest.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.carbpol.2019.115084>.

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